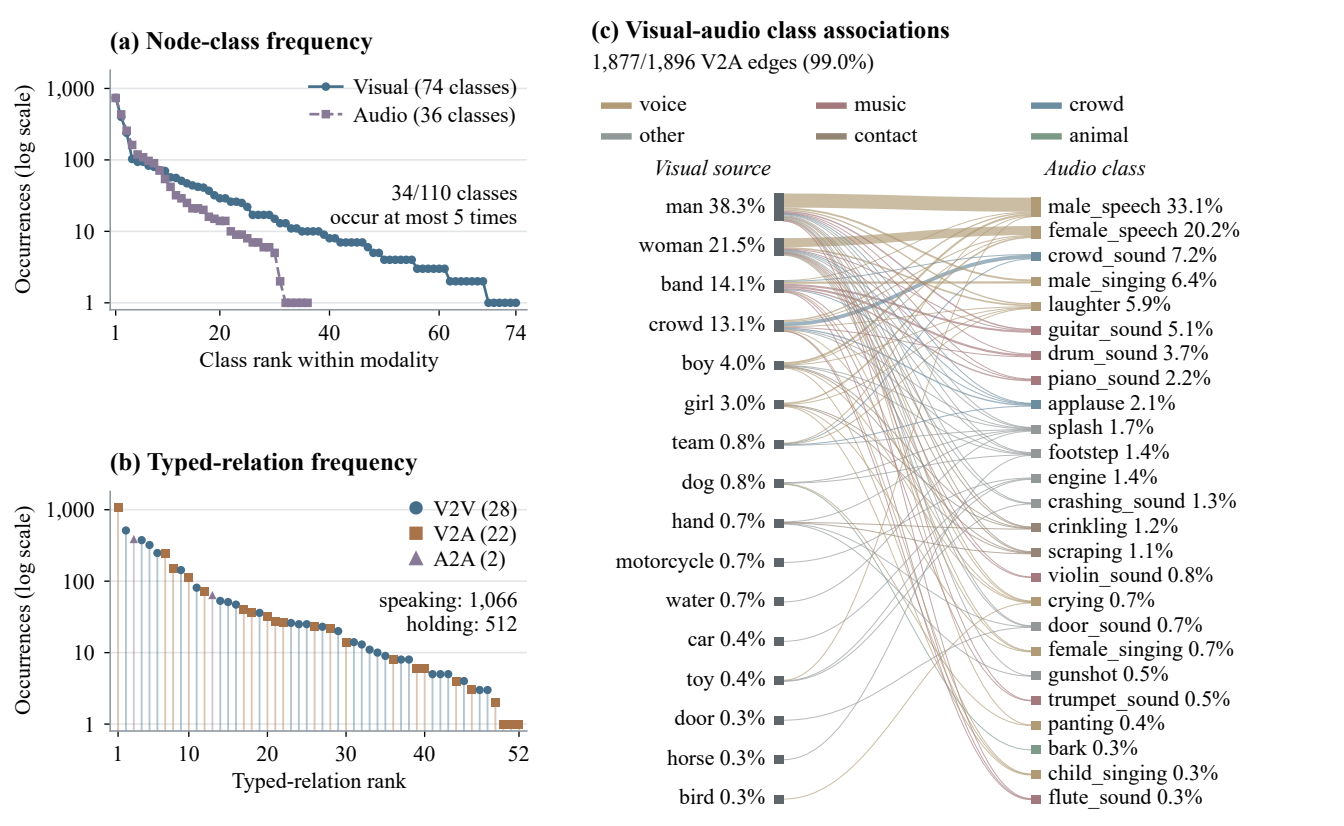

(a) Node-class frequency
Occurrences (log scale)
1,000
100
10
1
Visual (74 classes)
Audio (36 classes)
34/110 classes
occur at most 5 times
1
20
40
60
74
Class rank within modality
(b) Typed-relation frequency
Occurrences (log scale)
1,000
100
10
1
V2V (28)
V2A (22)
A2A (2)
speaking: 1,066
holding: 512
1
10
20
30
40
52
Typed-relation rank
(c) Visual-audio class associations
1,877/1,896 V2A edges (99.0%)
voice
music
crowd
other
contact
animal
Visual source
Audio class
man 38.3%
woman 21.5%
band 14.1%
crowd 13.1%
boy 4.0%
girl 3.0%
team 0.8%
dog 0.8%
hand 0.7%
motorcycle 0.7%
water 0.7%
car 0.4%
toy 0.4%
door 0.3%
horse 0.3%
bird 0.3%
male_speech 33.1%
female_speech 20.2%
crowd_sound 7.2%
male_singing 6.4%
laughter 5.9%
guitar_sound 5.1%
drum_sound 3.7%
piano_sound 2.2%
applause 2.1%
splash 1.7%
footstep 1.4%
engine 1.4%
crashing_sound 1.3%
crinkling 1.2%
scraping 1.1%
violin_sound 0.8%
crying 0.7%
door_sound 0.7%
female_singing 0.7%
gunshot 0.5%
trumpet_sound 0.5%
panting 0.4%
bark 0.3%
child_singing 0.3%
flute_sound 0.3%